Autonomous Drone for Dynamic Smoke Plume Tracking

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Project Website

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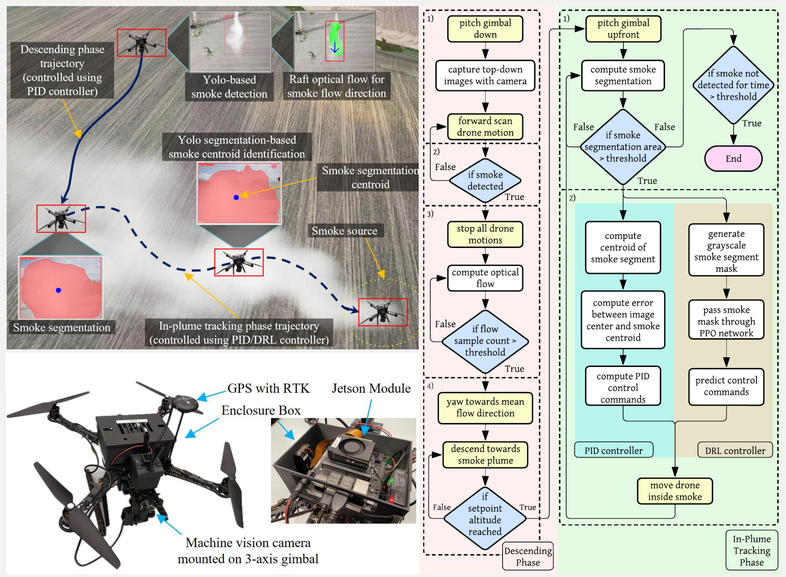
SAFL

Flow Field Imaging Lab

Introduction

- **Significance of Atmospheric Particle Transport**
  - Critical for environmental monitoring, climate modeling, and assessing public health [1-3].
  - Key sources include wildfires, volcanic eruptions, airborne sand and dust [4-6].
- **Multiscale Nature of Particle Transport**
  - Involves micro-sized particles to kilometer-scale plume dispersion events [7], influenced by wind, particles' morphology and flow conditions [8].
- **Challenges in Measurement**
  - Major gap remains in field-based, real-time high-resolution data collection tools.
  - **Objective:** Develop an autonomous drone system that detects and tracks dynamic plumes under unsteady wind conditions **– enabling intelligent vision-based navigation inside the smoke plume for precise data acquisition.**

Methodology

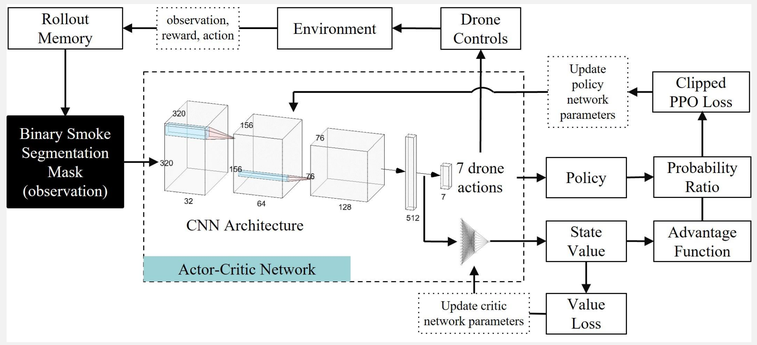


- **Algorithm Overview**
  - **Descending Phase:**
    - Yolo-based smoke detection.
    - RAFT optical flow for smoke flow direction.
    - Drone descends into smoke using PID (Proportional-Integral-Derivative) controller.
  - **In-plume Tracking Phase:**
    - Yolo-based smoke segmentation.
    - Drone tracks dynamic smoke plume using either PID or DRL (Deep Reinforcement Learning)-based controller.
- **Hardware & Software**
  - **Onboard Edge Computing**: Jetson Orin Nano with ROS Noetic enables real-time inference and backbone for autonomous drone control.
  - **Navigation**: Jetson Orin Nano with MAVROS ensures reliable MAVLink communication with the drone's Pixhawk flight controller.
  - **Imaging System**: 12MP ArduCam on a 3-axis gimbal enables visual top-down and in-plume smoke tracking.

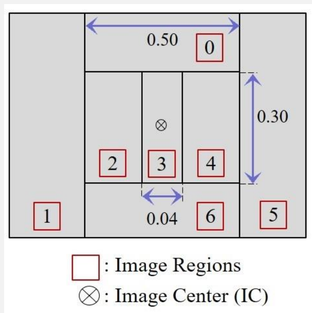
Methodology

- **DRL-based Drone Control Algorithm:**

  Used Proximal Policy Optimization (PPO) from Stable Baselines3 with Gym environments for robust DRL-based drone control.
  - **State Representation:** Binary images of smoke segmentation masks (320x320 pixels).
  - **Actor-Critic Network:**

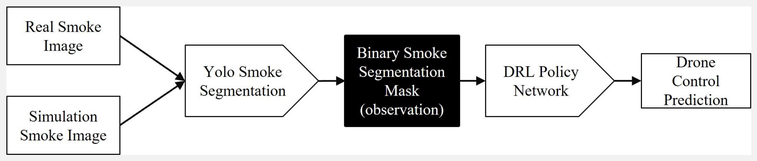


  - **Discrete Action Space:** 7 predefined velocity commands (up, down, left, hard left, right, hard right and straight) mapped to drone movements.
  - **Reward Function:**



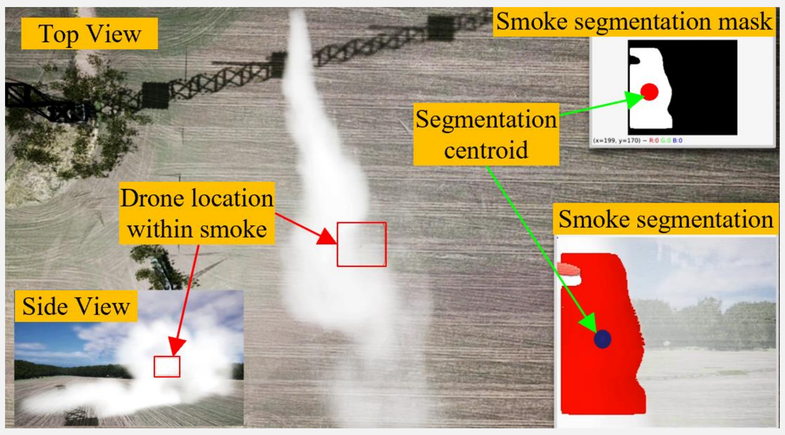
X_s in 0 or 6:
if [X_s= pred]: R=+1 *else:* -1
X_s in [1-5]:
if [pred = 0 or 6]: R= -1
else if [X_s= pred]: R=+1
else if [$X_s \approx$ pred]: R=+0.5 or 0 *else:* -1
1) X_s=1: *if* [pred=2]: R=+0.5
else if [pred=3]: R=0 *else:* -1
2) X_s=2: *if* [pred=1 or 3]: R=+0.5 *else:* -1
3) X_s=3: *if* [pred=2 or 4]: R=+0.5 *else:* -1
4) X_s=4 and X_s=5: R is symmetrical as X_s=2 and X_s=1 about IC respectively

  - **Training DRL Controller:** In simulation with realistic smoke dynamics to learn from various smoke patterns.
  - **Inference and Sim2Real Transfer:**

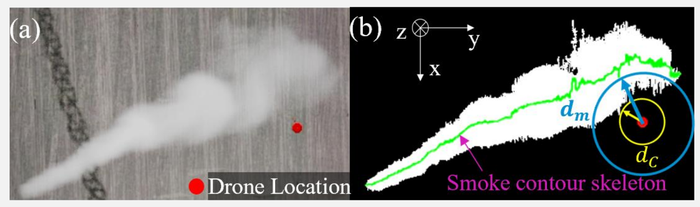


Simulation Environment

- Developed in Unreal Engine (UE) 5.1.1 replicating the Eolos Field Station for high-fidelity tests.
- UE with Niagara Plugin simulates dynamic smoke behavior under variable wind conditions.
- PX4 SITL + AirSim + ROS (via WSL2) enables realistic MAVLink-based communication for simulating realistic autonomous control.



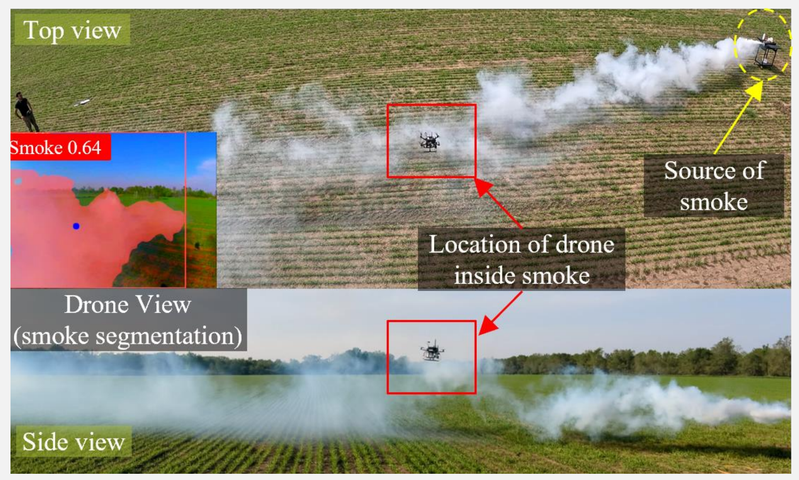
Simulation Assessment



- **Controlled Test Scenarios**: Four wind-smoke scenarios - Steady (S), Unsteady Low-Frequency (UL), Unsteady High-Frequency (UH), and 3D Fluctuations (U3D)] - were simulated for evaluation.
- **Evaluation Metrics**: 5 custom metrics used to quantify tracking accuracy – mean ($\tilde{\mu}_m$, $\tilde{\mu}_c$) and max ($\tilde{d}_{m,max}$, $\tilde{d}_{c,max}$) deviation from the plume mean line and plume contour, and percentage flight time spent inside the plume ($\tilde{t}_R$).
- **DRL vs. PID Performance**: **DRL outperforms PID in unsteady flows, staying inside the plume longer and at closer proximity to the plume.**

		$\tilde{\mu}_m$(%)	$\tilde{d}_{m,max}$(%)	$\tilde{\mu}_c$(%)	$\tilde{d}_{c,max}$(%)	$\tilde{t}_R$ (%)
S	PID	1.6±0.2	7.5±2.1	1.4±0.6	2.9±0.7	95.1±2.5
	DRL	1.4±0.2	7.0±2.3	1.8±1.2	4.0±3.3	94.1±1.4
UL	PID	4.0±0.4	11.9±2.2	2.5±1.3	6.6±3.1	87.1±2.2
	DRL	1.8±0.5	10.1±2.3	2.5±1.1	8.4±3.3	86.9±1.8
UH	PID	7.2±1.5	28.1±10.9	5.3±3.8	18.6±7.4	69.4±4.7
	DRL	5.4±1.1	26.0±8.4	4.6±3.1	12.3±6.5	85.0±6.4
U3D	PID	4.1±0.6	15.5±6.5	4.5±2.6	12.8±5.9	79.5±2.3
	DRL	1.9±0.5	10.6±4.4	1.5±2.1	4.3±4.3	95.0±4.1

Field Demonstration



	$\tilde{\mu}_m$(%)	$\tilde{d}_{m,max}$(%)	$\tilde{\mu}_c$(%)	$\tilde{d}_{c,max}$(%)	$\tilde{t}_R$(%)
PID	6.4	21.0	3.6	10.0	71.2
DRL	8.1	27.8	4.1	12.2	72.7

- **Although varying conditions prevent direct comparison, DRL achieved longer time in plume while PID maintained closer proximity.**

Conclusion & Discussion

- We present an autonomous drone system that uses PID and DRL controllers to track dynamic smoke plumes in real-world, unsteady atmospheric conditions.
- DRL excels in complex environments, maintaining better plume coverage in both simulation and field.
- This system has broad applications in wildfire monitoring, and air quality assessment with future work focused on improving generalization and robustness under extreme conditions.

Acknowledgement

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